

Traceability Information

Overview

This document maps each perception/object detection requirement to the design, implementation, and verification work completed for the IGVC senior design project.

Quick Map

- PER.FR.1 -> YOLOv8 pipeline -> trained model -> video-based testing
- PER.FR.2 -> detection output format -> inference output -> output review
- SW.R2 -> camera/video input flow -> recorded frames -> video-based testing
- SW.R4 -> YOLOv8 train + infer -> trained model -> performance evaluation
- PER.CON.1 -> outdoor data coverage -> labeled dataset -> lighting-condition tests

Traceability Matrix

Req ID	Requirement (short)	Design	Implementation	Verification
PER.FR.1	Detect barrels, potholes, signs	YOLOv8 pipeline	Trained model + labeled data	Tested on recorded video data
PER.FR.2	Provide outputs for integration	Output format defined	YOLOv8 inference output	Output review for usability
SW.R2	Camera/video input used	Input pipeline	Recorded frames as input	Tested using recorded video data
SW.R4	YOLOv8 real-time detection	YOLOv8 train + infer	Trained model	Model testing and performance eval
PER.CON.1	Lighting robustness	Outdoor-style dataset	Labeled dataset + training	Tested on varied lighting data

Notes

My work focused on the perception/object detection portion of the IGVC project, including dataset gathering, labeling, YOLOv8 training, and model testing.

The full system integration and navigation behavior depend on using the object detection outputs produced by the perception system.